Sao Cheng Tai

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MEng Computing (AI & ML) student at Imperial College London, strong in computer vision, machine learning and software engineering. London AI x Science Hackathon finalist; fine-tuned MedSAM2 with LoRA (0.969 Dice), built YOLO pipelines, web apps and robots, led teams of up to 16, and won two national AI Gold Awards. Passionate about collaborating with engineers and scientists on AI and computer vision software with real-world impact.

EDUCATION

Imperial College London *2026 – now*
MEng Computing (Artificial Intelligence and Machine Learning)

Yuet Wah College, Macau *2020 – 2026*
- International A Levels: 4A* (Mathematics, Further Mathematics, Physics, Chemistry)

SELECTED HACKATHON & RESEARCH EXPERIENCE

London AI x Science Hackathon – *Finalist, Polaron Materials Manufacturing Track (team of five)* *2026*
- Ranked in the top 8 of 50+ submissions (top 25 of 1,200+ applicants) against a field made up mainly of PhD and master's students, researchers, engineers and scientists.
- Team built Sentinel, a quality-control pipeline that classifies battery electrode microstructures in SEM images using U-Net segmentation, DINOv2 and a multi-agentic data explainer
- Designed and independently built the multi-agent explainer: analyst, materials-specialist and writer agents turn model outputs and scientific papers into cited, traceable explanations.
- Implemented Corrective RAG (CRAG): retrieved passages are graded for relevance, and weak results trigger query rewriting and a wider search, so only high-quality evidence is used.

Fast & Light: Low-Rank Adaptation for Medical Imaging – *Student Researcher (team of six)* *2025 – 2026*
- Fine-tuned MedSAM2, a SAM2-based medical segmentation model, with LoRA on 1.3% of its parameters, reaching 0.969 Dice in 72 minutes and cutting training time and cost by over 90%.
- Designed a dual-prompt fusion strategy using automated wide and central bounding boxes to handle anatomical challenges such as cardiac obscuration in chest X-rays.

SELECTED PROJECTS

Post-Disaster Debris Detection System – *Project Manager & Software Lead (team of six)* *2026*
- Co-developed a capstone prototype combining YOLOv12 detection with BeiDou satellite positioning to map post-disaster debris and fallen-tree hotspots in Macau.
- Led dataset curation and augmentation (Roboflow), trained YOLOv12 to 86.8% mAP, and built a real-time pipeline that turns imagery into public hotspot maps for clean-up decisions.

Intelligent Inspection Robot – *Individual project; National Gold Award* *2025*
- Built an autonomous robot car from scratch to navigate complex lanes, signs and obstacles.
- Implemented a real-time vision pipeline (denoising, adaptive thresholding, edge and object detection) to estimate lane offset for a hand-tuned PID motor controller.

Adaptive Smart Traffic Light System – *Project Manager & Software Lead* *2024*
- Led a 16-member team for three months to prototype a system in which YOLOv8 detections and scraped weather data feed an SVM that sets signal states via Arduino control.

HONOURS & AWARDS

- **Gold Award ×2**, National Youth AI Innovation Challenge (Inspection Robot; Smart Farm) *2025, 2023*
- **First Prize ×2**, Macau AI Challenge (1st and 3rd editions) *2025, 2023*
- **First Runner-Up ×2**, Youth Science & Technology Village (Engineering; Biology & Chemistry) *2026*
- **Macau PhD Think Tank High-Tech Scholarship** *2026*
- **High Distinction**, SDG Innovation Hackathon (Macau Selection Competition) *2025*
- **Award of Excellence**, Greater Bay Area Youth AI and Cybersecurity Challenge *2024*

TECHNICAL SKILLS

Computer Vision: image segmentation (e.g. U-Net, MedSAM2), object detection (e.g. YOLO), image preprocessing (denoising, adaptive thresholding, edge detection), augmentation, DINOv2; SEM and X-ray imaging
CV Tools: OpenCV, scikit-image, torchvision, Pillow, Ultralytics, MediaPipe, Roboflow
Scientific Computing & Data Science: NumPy, SciPy, pandas, scikit-learn, XGBoost, Matplotlib, Seaborn, Plotly; data curation, hyperparameter tuning, web scraping
Programming & AI: Python, C++, Arduino, PyTorch, Streamlit; multi-agent systems, RAG (e.g. CRAG)